

Luke M. Doudna

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Texas A&M University

B.S. in Computer Science, Minor in Mathematics

Relevant Coursework: Data Structures & Algorithms, Algorithm Design & Analysis, Computer Organization; In

Progress: Machine Learning, Software Engineering

College Station, TX

GPA: 3.63 | Expected Dec. 2027

TECHNICAL SKILLS

Languages: Python, Java, C/C++, C#, JavaScript/TypeScript, HTML/CSS, SQL

Frameworks & Libraries: React, Next.js, Node.js, Express, TensorFlow, PyTorch, D3.js, Tailwind CSS

Developer Tools: Git, Linux, Docker, Vercel, Firebase, Postman, Power Automate

Cloud & Platforms: AWS, Google Cloud, Azure DevOps, SQLite, MongoDB, REST APIs, OpenAPI/Swagger

EXPERIENCE

Hynes & Khater

Houston, TX

Software Engineering Intern

Jun. 2026 – Present

- Built PASS Travel as primary engineer on a small team, a full-stack app (React, Express, SQLite) for airline employees to enroll and manage dependent travelers for travel benefits, replacing a legacy system.
- Implemented role-based access control separating employee self-service from People Team (HR) approval workflows, built the full request lifecycle (submit, review, comment, approve, resubmit), and integrated an external employee API for traveler and profile synchronization.
- Built an AI-driven QA pipeline converting Azure DevOps user stories into executable Postman API tests using Copilot Studio and Power Automate, saving over 10 hours of manual test authoring in its first week of deployment.
- Improved generated-test accuracy by grounding the AI workflow in OpenAPI/Redocly API documentation, automating test-artifact storage and upload across Azure DevOps, SharePoint, and the Postman API.

GTI Geophysical Technology Inc.

Houston, TX

Software Engineering Intern

Summers 2023 & 2025

- Developed and tested firmware and software for geophysical measurement instruments, improving measurement accuracy and reliability.
- Automated a previously manual data-processing and analysis workflow with Python scripts, cutting turnaround from 2 hours to minutes per run.
- Built Windows-based C++ and C# applications for RFID reader integration, ensuring stable real-time connectivity and tag management.

PROJECTS

CodeAgent – Autonomous Coding Agent | *Python, Anthropic API, Docker*

2026

- Built an autonomous agent that plans and executes multi-step coding tasks through the Anthropic API.
- Implemented an iterative plan-act loop where the agent reads and edits files, runs code, and adjusts based on execution results until the task completes.
- Isolated code execution in Docker sandboxes for safe, reproducible runs across tasks.

Sift – Client-Side Search Engine | *Python, TF-IDF, Next.js, Vercel*

2026

- Built a responsive search engine using TF-IDF ranking that processes and indexes documents entirely client-side.
- Implemented tokenization, stemming, and relevance scoring from scratch without external NLP libraries.

Chess.com Analytics Dashboard | *JavaScript, D3.js, Chess.com API, Vercel*

2026

- Developed an interactive dashboard visualizing a player's Chess.com history and trends: rating progression, win/loss by opening, and time-control analysis.
- Integrated the Chess.com API to fetch and process game data, rendering visualizations with D3.js.

Song Wars – Multiplayer Music Game | *Firebase, Spotify API, Deezer*

2025

- Built a real-time multiplayer music game integrating the Spotify API and Deezer preview clips.
- Used Firebase for real-time game state, matchmaking, and player synchronization; deployed on Netlify.

LEADERSHIP & INVOLVEMENT

Freshman Leaders on Campus (FLOC)

College Station, TX

Staff

2024 – 2026

- Selected as returning staff to mentor first-year students through leadership workshops and service projects.
- Tutored middle and high school students in math (algebra, geometry), physics, and SAT prep.
- Competed in TAMUhack, Citadel Terminal (algorithmic trading competition), and a UT Austin coding competition.

Certification: AI & Machine Learning Certification (in-person, MIT/Boston University)